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PRIMARY RESEARCH PAPER

Drivers of metabolism and decomposition in forested Lake Superior tributary streams

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Abstract Ecosystem respiration (ER) and decomposition are fundamental carbon cycling processes in streams. Decomposition and ER may be controlled by similar environmental factors, as decomposition is carried out by heterotrophic bacteria, fungi, and macroinvertebrates, whose metabolic activity is a major contributor to ER. However, ER also includes respiration by autotrophs, and their respiration will be affected by factors that influence gross primary production (GPP). We quantified decomposition, GPP, and ER and estimated autotrophic respiration (AR) and heterotrophic respiration (HR) at eight sites in four forested streams in the Upper Peninsula of Michigan. These streams spanned gradients of canopy cover and dissolved organic carbon (DOC) concentrations, which we predicted would differentially affect activity due to changes in light and carbon availability. We also explored relationships between ER, AR, HR, and decomposition with other environmental drivers like temperature and nutrient availability. ER was related to TP concentrations but not DOC or canopy cover, and AR and HR were not related to any environmental drivers. Decomposition was related to canopy cover and dissolved nutrient concentrations, but not DOC. While our results did not demonstrate direct relationships between canopy cover or DOC concentration and respiration rates, we did find a wide variation in the fraction of respiration carried out by autotrophs relative to other studies, and that decomposition rates were comparable across study sites despite differences in GPP and ER. Together, these results suggest a need to improve methods and estimates of AR and HR in forested streams to integrate these important aquatic ecosystems into global climate models.

Keywords Metabolism · Autotrophic respiration · Heterotrophic respiration · Upper Peninsula of Michigan · Northern hardwood forest streams

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Terrestrial environments control light availability and supply dissolved and particulate organic carbon (C) to rivers and streams (Webster & Meyer, 1997; Battin et al., 2009), fueling the in-stream processes of metabolism and decomposition (Hynes, 1975; Vannote et al., 1980; Vachon et al., 2020).

Metabolism is the sum of gross primary production (GPP), the transformation of inorganic to organic C using sunlight as an energy source, and ecosystem respiration (ER), the breakdown of organic C into an inorganic form to release stored energy. Most stream ecosystems are heterotrophic, where rates of ER exceed rates of GPP, often due to canopy cover limiting light input (Vannote et al., 1980; Mulholland et al., 2001). Streams also receive organic C subsidies from the surrounding watershed, commonly referred to as allochthonous C (Webster & Meyer, 1997; Marcarelli et al., 2011). Allochthonous particulate C includes leaf and other terrestrial litter, which is broken down through a combination of physical processing and decomposition by heterotrophic organisms (macroinvertebrates, bacteria, and fungi), in turn fueling rates of ER. In forested ecosystems, light exposure to the water surface increases and the relative availability of allochthonous inputs decreases as stream width increases. This can shift the balance between GPP and ER and increase the relative importance of autotrophic processes in downstream reaches (Vannote et al., 1980; Bernhardt et al., 2022).

In rivers in Northern Michigan and throughout the upper Midwestern United States, dissolved organic C (DOC) concentrations tend to be high due to widespread wetland inputs. DOC concentrations range from 2.14 to 9.30 mg C L^{-1} in streams of the Upper Peninsula of Michigan (Coble et al., 2016a) which is higher than 1–5 mg C L^{-1} reported from headwater streams without drainage from wetlands or organic-rich soils (Findlay & Parr, 2017). High concentrations of DOC limit light penetration in surface waters (de Wit et al., 2016) and increase surface water temperatures in lakes due to decreased water clarity and increased light absorption (Read & Rose, 2013; Williamson et al., 2015; Pilla et al., 2018). The effects of DOC on in-stream metabolic processes are complex because DOC can be an energy source for heterotrophic bacteria, but it can reduce light penetration, which can lead to lower overall GPP (Godwin et al., 2014). Solomon et al. (2013) examined the effect of DOC on ER in lakes and found no significant relationship, which they hypothesized was because ER is actually composed of autotrophic respiration (AR) and heterotrophic respiration (HR), and the differing characteristics of autotrophs and heterotrophs may lead to different responses to common environmental drivers.

AR is respiration that is performed by primary producers, and rates of AR should be closely tied to rates of GPP. Therefore, AR should be affected by the factors that have demonstrated influence on primary producers, such as nutrient concentration and light availability (Mulholland et al., 2001). Lower light availability due to either increases in canopy cover or increases in DOC concentrations could therefore decrease AR. GPP also tends to increase with water temperature (Griffiths et al., 2013; Demars et al., 2016), and water temperature can be decreased by increased canopy cover and by decreased DOC concentrations (LeBlanc et al., 1997; Johnson, 2004). Finally, nutrient availability is an important influence on rates of GPP and AR, as primary production can be limited by the availability of inorganic forms of nitrogen and phosphorus (Griffiths et al., 2013).

HR is respiration that is performed by all organisms that do not fix their own C such as bacteria, fungi, and invertebrates. Therefore, HR and AR may respond similarly to some environmental factors but differently to others, and HR may be influenced by similar factors as those that control decomposition rates (Benstead et al., 2009; Ferreira et al., 2015). Similar to AR, temperature is a primary control on decomposition, with higher water temperatures leading to higher rates of decomposition (Martínez et al., 2014; Ferreira et al., 2020) and respiration (Jankowski et al., 2014). Decomposition and HR can also be increased by the supply of inorganic nutrients (Rosemond et al., 2015; Costello et al., 2022), but decomposition also leads to remineralization of nutrients that can then stimulate GPP and ER (Parton et al., 2007). In contrast to AR, HR may increase with higher DOC concentration since heterotrophs could use the increased organic C to fuel their own processes, assuming that the DOC is bioavailable, which may not always be true (Wymore et al., 2015; Coble et al., 2016b). HR should also be influenced by factors that affect the availability of both particulate and dissolved organic C, such as canopy cover, acidity, and discharge (Tank et al., 2010; Demars, 2019). Finally, HR and decomposition may also be affected by rates of GPP, as labile C exuded by autotrophs may positively prime microbes to carry out decomposition (Halvorson et al., 2019). The responses of AR and HR to changes in environmental conditions may

therefore be complicated, requiring quantification of both processes in the field.

It is important to understand which environmental factors influence these processes as many of these factors are likely to change due to anthropogenic activity. For example, understanding how increases in DOC concentrations affect both respiration and decomposition can improve predictions of how river ecosystems may respond to increasing DOC concentrations and increased temperatures from climate change (Roulet & Moore, 2006). Nutrient pollution from point and non-point source inputs can accelerate the loss of terrestrial leaf litter from streams (Rosemond et al., 2015), and global patterns in afforestation and deforestation may alter the riparian inputs of allochthonous materials to streams (Riis et al., 2020). We designed two studies to examine how canopy cover and DOC concentrations affected metabolism and decomposition rates in Lake Superior tributary streams located in the vicinity of the Keweenaw Peninsula in the Upper Peninsula of Michigan. The first, hereafter the "Longitudinal Study," examined the variation in ER, HR, AR, and decomposition along the Pilgrim River with sites distributed from the headwaters to just upstream of where the river joins the Keweenaw Waterway. We hypothesized that ER (AR + HR) rates would be higher in downstream sites than in headwater sites, due to an increase in rates of GPP and AR with increasing light availability due to increased stream width and decreased canopy cover at downstream sites. We also hypothesized that decomposition and HR rates would be lower in downstream sites due to lower availability of particulate organic C from litter inputs, and therefore the downstream sites would host proportionally fewer organisms specialized for decomposition. The second study, hereafter the "DOC Gradient Study," examined the variation and environmental predictors of metabolism and decomposition among four streams with different DOC concentrations. We hypothesized that ER rates would not be affected by DOC concentrations, due to the offsetting responses of HR and AR. We hypothesized that higher DOC concentrations would stimulate HR but also result in more browning of the water (de Wit et al., 2016) which would limit light availability, causing AR to decrease. Finally, we used data from both studies together to evaluate how water temperature and nutrient concentrations, along with canopy cover and DOC concentrations, were related to rates of ER, decomposition, AR, and HR.

Methods

Study area

For the Longitudinal Study, we chose five sites along the Pilgrim River, located in Houghton County, Michigan (Fig. 1a). The watershed is 62.9 km^2 with a total stream length of 21.1 km and an annual discharge of 0.98 $m^3 s^{-1}$ (Houghton Keweenaw Conservation District, 2012; U. S. Geological Survey, 2016). We sampled five sites total, and sites are named according to their distance from the river mouth, where the Pilgrim River enters the Keweenaw Waterway: Pilgrim 1, Pilgrim 6 (co-located with USGS gage 04043016), Pilgrim 14, Pilgrim 15 (located on North Fork Pilgrim River), and Pilgrim 16 (located on South Fork Pilgrim River). The terrestrial vegetation is dominated by hardwood sugar maple and yellow birch forest (Houghton Keweenaw Conservation District, 2012), with canopy cover ranging from 3.5 to 96.4% among sites (Table 1). Notably, headwater site Pilgrim 16 had an almost entirely closed canopy (96.4% closed), while the other headwater site Pilgrim 15 was located in a wetland and had an almost entirely open canopy (3.5% closed). We sampled sites monthly from May 2019 until August 2019.

The DOC Gradient Study included Pilgrim 1 as well as sites in three other rivers: the Trap Rock, the Silver, and the Tobacco (Fig. 1b). These sites were selected due to proximity to a USGS gage and previous data that suggested a range in DOC concentrations (A. M. Marcarelli, unpublished data). The Trap Rock site was located 6 km upstream from the river mouth entering Torch Lake, at USGS gage 04043050 near Lake Linden, Michigan in Houghton County. The Trap Rock has a watershed area of 111.37 km^2, average annual discharge of 1.29 $m^3 s^{-1}$, and land cover comprised of deciduous forest, mixed forest, and woody wetlands (U. S. Geological Survey, 2016; Michigan Department of Environmental Quality, 2022). The Silver site was located 4 km upstream of the mouth of the river entering Huron Bay of Lake Superior, at USGS gage 04043150 near L'Anse, Michigan in Baraga County. The watershed of the Silver is 92%

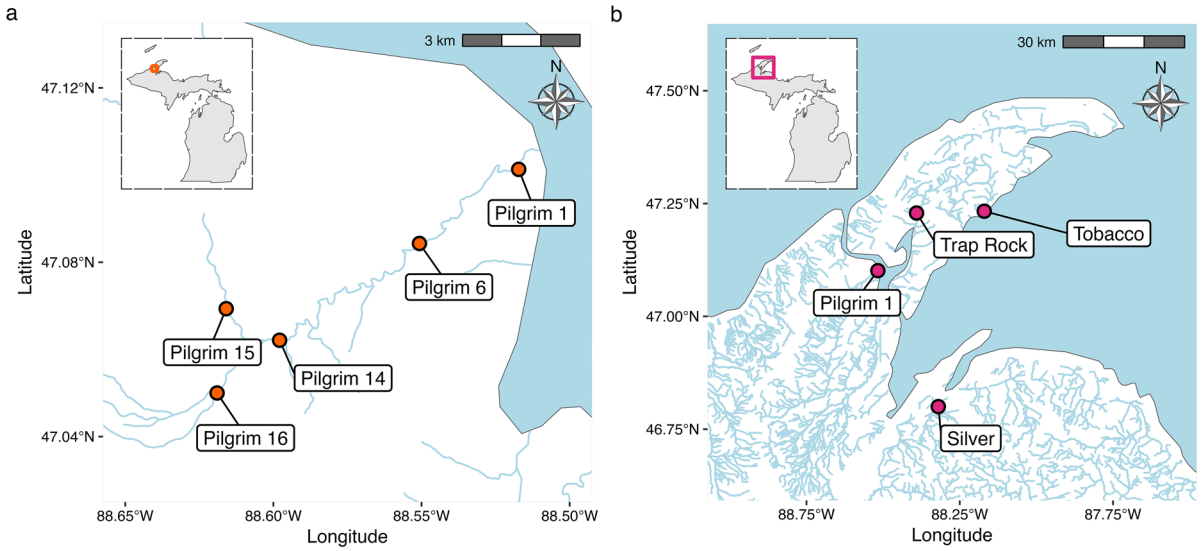

Fig. 1 The distribution of study sites in the Upper Peninsula of Michigan. The Longitudinal Study sites **a** were distributed along the Pilgrim River, while the DOC Gradient study sites **b** were located within Baraga, Houghton, and Keweenaw counties. Insets show the location and extent of the main maps within the state of Michigan

hardwood forest, has an area of 178.71 km^2, and the average annual discharge of the river is 2.44 m^3 s^{-1} (Weaver et al., 2010; U. S. Geological Survey, 2016). The Tobacco site was located near Gay, Michigan in Keweenaw County with land cover dominated by hardwood forests with some scrub palustrine wetlands (Keweenaw County Planning Commission, 2017). The study site was located 2 km upstream from the mouth of the river, and the area of the surrounding watershed was 137.15 km^2 (Michigan Department of Environmental Quality, 2025). We measured environmental characteristics, such as water chemistry, canopy cover, and stream width once every two weeks from May 2020 until October 2020 (summarized in Table 1).

Environmental characteristics

We characterized physical and chemical conditions at each study site, using consistent methods for both studies. We measured canopy cover using a densiometer. We collected stream water from each site and filtered using Millipore 0.45 µm nitrocellulose membrane filters into acid washed plastic Nalgene bottles for measurement of dissolved water chemistry. We froze samples until analysis was performed for nitrate (NO_3^-), ammonium (NH_4^+), total dissolved phosphorus (TDP), dissolved organic carbon (DOC), and total dissolved nitrogen (TN). We collected water chemistry samples and canopy cover measurements once a month for the duration of the Longitudinal Study, and once every two weeks for the duration of the DOC Gradient Study, for a total of 2 and 9 sampling dates, respectively. The DOC and TN concentrations were measured using a Shimadzu TOC-VCSN analyzer with a total N module TNM-1 (Shimadzu Scientific Instruments, Columbia, Maryland). The NH_4^+ concentrations were measured using a fluorometric procedure (Holmes et al., 1999; Taylor et al., 2007) on a Turner Aquafluor (Turner Designs, Palo Alto California). The TDP concentrations were measured using acid-persulfate digestion followed by molybdenum-antimony colorimetric determination methods with a Thermo Scientific 10 s UV–Vis spectrophotometer (Nydahl, 1978; Valderrama, 1981; Ameel et al., 1993; Water Environmental Federation & American Public Health Association, 2005). NO_3^- was also analyzed on a SEAL AQ2 discrete analyzer (SEAL Analytical, Mequon, Wisconsin) based on USEPA method 353.2 revision 2.0 (O'Dell, 1993) and APHA method 4500 NO_3^- (Water Environmental

Table 1 Mean (± standard deviation) environmental characteristics for each site

	Longitudinal Study (2019)					DOC Gradient Study (2020)			
Site	Pilgrim 16	Pilgrim 15	Pilgrim 14	Pilgrim 6	Pilgrim 1	Pilgrim 1	Tobacco	Silver	Trap Rock
Latitude	47.0500	47.0693	47.0621	47.0843	47.1013	47.1013	47.2330	46.8041	47.2290
Longitude	− 88.6190	− 88.6169	− 88.5979	− 88.5507	− 88.5172	− 88.5172	− 88.1700	− 88.3169	− 88.3853
USGS gage	–	–	–	04043016	–	–	–	04043150	04043050
Study start date	14 May	14 May	11 May	20 May	14 May	6 Jun	29 Jul	6 Jun	6 Jun
Study end date	6 Sep	6 Sep	6 Sep	6 Sep	6 Sep	20 Oct	20 Oct	20 Oct	20 Oct
Width (m)	2.8**	3.3**	5.5**	8.8**	7.5**	8.3**	9.5 ± 0.18	14.6**	10.5**
k600 (d^{-1})	10.47 ± 4.05	2.56 ± 3.12	23.35 ± 4.01	22.35 ± 1.00	18.80 ± 2.88	9.74 ± 5.20	9.57 ± 2.50	25.51 ± 14.10	20.13 ± 5.15
Areal extent (km)	*	*	1.6	1.5	0.8	6.5	4.5	2.5	2.6
DOC (mg C L^{-1})	5.35 ± 0.3	15.05 ± 3.49	7.15 ± 0.49	5.30 ± 0.39	6.35 ± 1.49	8.61 ± 7.13	16.63 ± 2.67	11.33 ± 3.99	6.66 ± 1.32
Canopy Cover (%)	96.4	3.5	90.9	13.4	19.7	11.7	6.0	27.4	9.4
Discharge (m^3 s^{-1})	0.10**	0.02**	0.21**	0.21**	0.38**	0.74 ± 9.62	0.51 ± 0.50	1.51 ± 1.76	0.79 ± 0.73
Temperature (°C)	12.75 ± 2.06	16.93 ± 3.28	14.31 ± 2.09	13.68 ± 2.43	13.14 ± 2.81	14.64 ± 4.27	14.88 ± 4.82	14.54 ± 4.93	13.87 ± 4.05
NH_4^+ (µg N L^{-1})	6.92 ± 1.06	6.95**	7.42 ± 0.61	9.19**	7.95 ± 3.14	8.16 ± 6.04	19.23 ± 10.05	9.15 ± 3.38	5.51 ± 2.54
TN (µg N L^{-1})	375 ± 21	390 ± 42	320 ± 37	280 ± 56	270 ± 77	360 ± 150	490 ± 100	325 ± 58	328 ± 44.5
NO_3^- (µg N L^{-1})	280 ± 46	*	230 ± 11	170 ± 35	140 ± 7	30 ± 40	19 ± 8	22 ± 15	33 ± 47
TDP (µg P L^{-1})	5.67 ± 5.33	4.64 ± 3.88	11.49 ± 2.90	14 ± 2.9	10 ± 4.4	9.38 ± 5.47	6.43 ± 0.72	7.83 ± 1.44	7.22 ± 3.13

Longitudinal Study sites were measured in 2019 and DOC Gradient Study sites were measured in 2020. The gas exchange rates (k_{600}) and areal extents for the single-station metabolism models are also reported for each site.

— = not applicable. * = not measured. ** = one measurement collected over study period.

Federation & American Public Health Association, 2005).

Metabolism modeling

In order to estimate rates of GPP and ER, we used a single-station open-water model based on changes in dissolved oxygen (Hall & Hotchkiss, 2017). We measured dissolved oxygen and temperature at 10-min intervals using a miniDOT dissolved oxygen (DO) sensor (PME Instruments, Vista CA) at each study site (see Online Supplementary Materials Figs. 1–4 for the full datasets). We deployed sensors from May 14 until September 6, 2019 for the Longitudinal Study and June 12 to October 20, 2020 for the DOC Gradient Study, except at the Tobacco where the sensor was deployed on July 29, 2020. We measured stream width as the distance between wetted banks at the location of the dissolved oxygen sensor. Stream discharge was either measured directly or obtained from nearby USGS gages (Table 1). During the Longitudinal Study, we measured discharge once at each site using a Marsh-McBirney Flo-Mate Model 2000 Portable Flowmeter (Hach, Loveland, CO). Unfortunately, the USGS discharge sensor at Pilgrim 6 (gage 04043016) was not operational during 2019 due to damage from a 1000-year flood event in June 2018. However, 2019 gage data from the nearby Trap Rock suggest that discharge was relatively stable during the study period (Online Supplementary Materials Fig. 5). During the DOC Gradient Study, discharge was obtained from USGS stream gages (Table 1) at the Trap Rock and Silver sites. The USGS station at Pilgrim 6 was redeployed

on June 9, 2020, and data from this gage were used for the Pilgrim 1 site after comparisons between measured discharge at Pilgrim 1 and gaged discharge at Pilgrim 6 suggested minimal differences (Nevorski & Marcarelli, 2022). At the Tobacco site, we measured discharge with the Marsh-McBirney flow meter every two weeks. A linear regression indicated that the discharge measured at the Tobacco site could be accurately predicted from discharge measured at the Pilgrim River USGS gage during the same time period (Tobacco Q = 0.703 × (Pilgrim 6 Q) + 0.0659, where Q = discharge in $m^3\ s^{-1}$; $R^2 = 0.95$, F = 64.12, df = 2, p = 0.02). We used this linear regression equation to estimate discharge for the Tobacco throughout the study period.

Using the gaged, measured, or estimated stream discharge, we then estimated average channel depth (Leopold & Maddock, 1953) at all sites with the equation:

$$z = cQ^f, \tag{1}$$

where c is the unit discharge coefficient (m), f is the depth–discharge relationship (unitless), Q is discharge ($m^3\ s^{-1}$), and z is depth (m). Guided by the exponential regression from the meta-analysis of depth–discharge relationships in Raymond et al. (2012), we set c = 0.409 and f = 0.294.

We used dissolved oxygen, temperature, stream depth, and discharge to estimate GPP and ER with Bayesian inverse modeling. We used the *streamMetabolizer* package for R (Appling et al., 2018; R Core Team, 2021) to apply the single-station metabolism equation (Eq. 2; Van De Bogert et al., 2007):

$$O_t = O_{t-\Delta t} + \frac{GPP}{z} \times \frac{PAR_t}{PAR_{total}} + \frac{ER}{z}\Delta t + K\Delta t\left(Osat_{t-\Delta t} - O_{t-\Delta t}\right), \tag{2}$$

where O_t is oxygen concentration (g O_2 m^{-3}) at time t, $O_{t\text{-}\Delta t}$ is oxygen concentration (g O_2 m^{-3}) at previous time t-Δt, GPP is gross primary production (g O_2 m^{-2} d^{-1}), z is average stream depth (m), PAR_t is photosynthetically active radiation at time t (μmol m^{-2} s^{-1}), PARtotal is sum photosynthetically active radiation within one day (μmol m^{-2} s^{-1}), ER is ecosystem respiration (g O_2 m^{-2} d^{-1}), Δt is the length of time between time t and time $t+1$ (d), K is the gas exchange rate across the air–water interface (d^{-1}), and $Osat_{t\text{-}\Delta t}$ is the saturated dissolved oxygen concentration under in-situ temperature and pressure at previous time t-Δt. We estimated PAR following the approach of *streamMetabolizer* (Appling et al., 2018), where maximum possible solar insolation at the earth's surface without canopy or cloud cover, a constant, is multiplied by the cosine of the site's zenith angle, which changes with latitude and time.

We fit the model using the default prior probabilities of GPP ~ N(μ = 3.1, SD = 6), ER ~ N(μ = − 7.1, SD = 7.1), and k_{600} ~ N(μ = log(12), SD = 1.32), where k_{600} is a temperature-normalized version of K. While direct measurements of gas exchange can improve model performance (Hall & Hotchkiss, 2017), due to time, weather, and constraints on fieldwork because of COVID-19 restrictions, we were unable to collect field measurements of K to help parameterize model fit. Using inverse modeling guided by prior probability distributions, the model uses O_2, water temperature, depth, and light to determine estimates of GPP, ER, and K for each full day of measurement. The model structure assumes that ER is constant throughout the day, and nighttime change in O_2 is attributable to only ER and K. As ER represents the consumption of O_2, we report the values as negative numbers, where greater |ER| represents greater respiration.

We calculated the areal extent of the stream reach integrated by metabolism measurements to assess whether physical conditions upstream of the sensor site were homogenous (Table 1; Odum, 1956). To ensure no overlap between Longitudinal Study sites, we compared the areal extent of Pilgrim River metabolism estimates to the distance between sampling site locations. Areal extent is the upstream distance integrated by metabolism measurements, which is estimated as three times the flow velocity (v, in m s^{-1}) divided by temperature-specific gas exchange (K, in s^{-1}; Table 1; Chapra & Di Toro, 1991). We estimated flow velocity using stream discharge, wetted width, and depth ($v = Q/A$, where A is width times depth). The estimated areal extents for the Pilgrim River sites ranged from 0.8 to 1.6 km, which is less than the distance between sites of ≥ 5 km (Table 1). For the sites in the DOC Gradient Study, all four streams had no

major discontinuities upstream of the study reach within the areal extent distance, so a single-station assumption was appropriate.

We evaluated metabolism models for equifinality, an indicator of poor model performance. Equifinality occurs if many different values of GPP, ER, and k_{600} could successfully be fit to the O_2 data, or if large values of GPP, ER, and k_{600} are just as likely to successfully fit the data as ecologically realistic values (Appling et al., 2018). To check for equifinality, we compared daily mean ER at each site to the site's daily mean k_{600}, as a significant linear relationship between these variables indicates equifinality (Appling et al., 2018). None of the DOC Gradient Study sites exhibited equifinality, but both headwater sites in the Longitudinal Study (Pilgrim 15 and 16) did exhibit equifinality. Therefore, we were only able to successfully model metabolism for 3 of the 5 Longitudinal Study sites (Pilgrim 1, Pilgrim 6, and Pilgrim 14).

Autotrophic and heterotrophic respiration

We used a quantile regression approach following the procedure of Hall & Beaulieu (2013) to estimate rates of AR using the *quantreg* package in R (Koenker, 2020). This approach uses the 90th percentile slope of quantile regression between ER and GPP multiplied by daily GPP to calculate AR (Eq. 3), as AR is hypothesized to be the minimum amount of ER on any day above the base HR. The equation that is used to estimate AR is as follows:

$$AR = AR_f \times GPP, \tag{3}$$

where AR_f is the slope of the 0.9 quantile regression between ER and GPP, representing the fraction of primary production that is immediately respired by autotrophs. As total ER should be comprised of AR and HR, HR can be estimated by subtracting AR from ER (Eq. 4).

$$HR = ER - AR \tag{4}$$

Because this is a statistical approach to estimate these different respiration rates, there are strict assumptions that must be met, which we assessed following the recommendations by Hall & Beaulieu (2013). We did not use any estimates where the confidence interval of the quantile regression slope was greater than 0.4 as Hall & Beaulieu (2013) considered any estimates with error was greater than that to not be too uncertain to be insightful. We also examined the correlation between HR and GPP, and we did not use any sites with a correlation of 0.3 or higher, as that was shown to decrease or increase the AR_f estimates by 0.15, depending on the sign of the correlation. Also, any sites that had low variation in GPP were found to have unrealistic estimations of AR rates (as in Hall & Beaulieu, 2013), so we excluded these sites from further analyses. Based on these criteria, AR and HR were estimated for the Silver, Pilgrim 6, and Pilgrim 1 (in both 2019 and 2020) sites, and could not be estimated for the Pilgrim 14, Tobacco, and Trap Rock sites.

Decomposition

We deployed cotton strip assays at each site following the procedure outlined in Tiegs et al. (2013). This method provides a standardized approach of measuring microbial decomposition relative to the more commonly used leaf packs (Boyero et al., 2016; Follstad Shah et al., 2017). Cotton strips respond similarly to environmental conditions like nutrient concentrations and temperature compared to leaf packs (Tiegs et al., 2013, 2019; Costello et al., 2022) but have fewer complicating factors like litter quality, nitrogen:phosphorus ratios, and lignin concentrations (Webster & Benfield, 1986; Güsewell & Gessner, 2009; Boyero et al., 2016). This method does not incorporate decomposition performed by invertebrates. We deployed cotton strips one time at each site and incubated them for 27–33 days. Cotton strips were deployed on Jul 2, 2019 for the Longitudinal Study and Sep 17, 2020 for the DOC Gradient Study. Cotton strips were attached to nylon cord by cable binders with 5 replicates per site. These strips were placed in alignment with the current and secured on the stream bed with cobble found within the stream. Upon collection, we washed the strips with 90% ethanol, then dried at 40 °C and placed in a desiccator to await tensile strength measurement. Lower tensile strength, or the amount of pulling force a material can receive until it breaks, should correspond to greater decomposition of the cotton fibers. We measured the tensile strength of the incubated strips, as well as control strips, by pulling at a rate of 2 cm min^{-1} to measure maximum tensile strength using an Instron

4206 load frame (Instrom, Norwood, MA) with an MTS Renew controller and 45.35 kg load cells (MTS Systems, Eden Prairie, MN). We calculated decomposition rates using Eq. 5, similar to Mancuso et al. (2022):

$$k_{decomp} = -\ln \frac{T_s}{T_{sc}} \times \frac{1}{days}, \quad (5)$$

where $k_{decomp.}$ is the decomposition rate, T_s is the maximum tensile strength of the incubated strips, T_{sc} is the mean tensile strength of the control strips, and days is the number of days the strips were incubated. Temperature has a large effect on decomposition rate and correcting for degree days allows the examination of decomposition in relation to factors other than temperature (Pozo et al., 2011). Therefore, we calculated k_{decomp} (Eq. 5) using both the number of days the strips were incubated (referred to as k_d, representing decomposition rate per day) or using the degree days (referred to as k_{dd}, representing decomposition rate per degree day). Degree days were calculated as the sum of mean daily water temperatures measured by miniDOT sensors that were deployed for metabolism estimates, divided by the number of sampling days (Pozo et al., 2011).

Statistical analysis

To infer if rates of GPP and ER were significantly different among Longitudinal or DOC Gradient Study sites during the duration of each study, we used a Kruskal–Wallis one-way analysis of variance with study days as replicates, followed by a Dunn's test with a Bonferroni correction to determine which sites differed. Because there were only two sites where we could estimate AR and HR in both the Longitudinal and DOC Gradient studies, we compared AR and HR rates between sites using a Dunn's test with a Bonferroni correction using the *PMCMRplus* package in R (Pohlert, 2024). We opted for these non-parametric tests because the metabolism data did not meet the normality and homoscedasticity assumptions of regular analysis of variance (ANOVA). Decomposition rates did meet those assumptions, so we used ANOVA followed by a Tukey HSD to examine if decomposition rates differed among sites in each study.

To determine which environmental characteristics were significantly related to rates of ER, we assembled a dataset using the nutrient grab-sampling and metabolism data from both studies on each grab-sampling date (n = 40 total site and date combinations). Predictors (n = 10) were DOC, TN, TDP, NH_4^+, NO_3^-, canopy cover, mean daily water temperature, maximum daily water temperature, k_d, and k_{dd}. Depth and discharge were not included as predictors because depth was used to model metabolism rates (Eq. 2), and we used discharge to estimate depth (Eq. 1). We then tested to see which environmental variables best predicted ER rates using partial least squares regression (PLSR) using the pls package in R (Liland et al., 2024). We chose PLSR due to its suitability for use with highly correlated or noisy predictors, including those with non-independent observations, as this analysis combines predictor variables into uncorrelated and independent orthogonal components (Wold et al., 2001). Before conducting the PLSR, we pre-processed data by log transforming to meet the assumptions of PLSR models (Kothawala et al., 2014), centering, scaling, and removing variables with near-zero variance. We determined near-zero variance by comparing the frequency of the most common value in a variable to the frequency of the second most common value. If the most common value accounted for 95% or more of the data, and the second most common value accounted for 5% or less of data (i.e., a cutoff frequency of 95/5), the variable was classified as having near-zero variance and excluded from the dataset, as its inclusion would not contribute additional explanatory value to the model. Afterward, we imputed any missing values using 5-nearest neighbors (56 values, 9.3% of dataset), and then removed any highly correlated predictors (cutoff value of 85%). After pre-processing, the dataset contained 40 observations of 7 predictor variables, which were canopy cover, DOC, TN, TDP, NH_4^+, NO_3^-, and mean daily water temperature.

To determine which environmental characteristics were significantly related to rates of decomposition, we created a subset of the original dataset that contained only sampling dates that coincided with the duration of cotton strip deployments (July 25-August 27, 2019 for the Longitudinal Study; September 17-October 20, 2020 for the DOC Gradient Study; n = 14 site and date combinations), and the predictors canopy cover, DOC, TN, TDP, NH_4^+, NO_3^-, depth,

GPP, and ER. This dataset was then pre-processed following the same steps as the ER dataset. After pre-processing there were 13 observations of 9 predictor variables, which were canopy cover, DOC, TN, TDP, NH_4^+, NO_3^-, GPP, and ER. There were 9 imputed missing values, which accounted for 3.2% of the dataset.

For both ER and decomposition, we assessed which predictors were most important by calculating the variable importance on the projection (VIP) scores for the PLSR. VIP scores greater than 2 were considered highly influential, those between 1 and 2 were considered moderately influential, and less than 1 as not influential on the model (Kothawala et al., 2014). We also evaluated the strength and direction of relationships between environmental variables, AR, and HR using Pearson correlation with a Bonferroni correction.

Results

Environmental characteristics

Consistent with the Longitudinal Study design, canopy cover varied among sites in this study, but water chemistry and other environmental factors did not. Canopy cover ranged from almost completely open to completely covered, with the strongest contrast between the two headwater sites (3.5% closed canopy at Pilgrim 15 vs. 96.4% closed canopy at Pilgrim 16). Canopy cover decreased as expected among the mainstem sites, with 90.9% closed canopy at Pilgrim 14, 13.4% at Pilgrim 6, and 19.7% at Pilgrim 1. The DOC concentrations, measured once a month from May 2019 until August 2019, mostly fell within the range of 5.35–7.15 mg C L^{-1}, although Pilgrim 15 had a concentration of 15.05 mg C L^{-1}. There was no clear trend in DOC concentration with the longitudinal gradient. NO_3^- and TDN generally decreased from headwaters to downstream, while NH_4^+ and TDP were similar among sites (Table 1).

Consistent with the DOC Gradient Study design, DOC concentrations increased from 6.7 mg C L^{-1} at the Trap Rock to 8.05 mg C L^{-1} at Pilgrim 1 and 11.3 mg C L^{-1} at the Silver, to a high of 16.63 mg C L^{-1} at the Tobacco (Table 1). Canopy cover was relatively similar at all sites (ranging from 6.0% to 27.4% closed), as was water temperature (ca. 14 °C on average). TDP was similar among sites, but the Tobacco had higher concentrations of NH_4^+, lower concentrations of NO_3^-, and higher concentrations of TN compared to the other sites, reflecting the influence of upstream wetlands at this site (Table 1).

Gross primary production and ecosystem respiration

In the Longitudinal Study, there was strong correlation between ER and k_{600} at Pilgrim 15 ($R^2 = 0.98$, $n = 115$ days) and 16 ($R^2 = 0.89$, $n = 115$ days; Online Supplementary Materials Fig. 6), which may indicate poor model performance or equifinality in the modeling analysis as described above (Appling et al., 2018). Therefore, the GPP and ER data from these two sites were left out of further analyses. Pilgrim 14, Pilgrim 6, and Pilgrim 1 had significantly different rates of GPP (Kruskal–Wallis test $\chi^2 = 153.09$, degrees of freedom (df) = 2, $p < 0.001$), and ER (Kruskal–Wallis $\chi^2 = 183.07$, df = 2, $p < 0.005$), with rates of both GPP and ER increasing from upstream to downstream (Fig. 2). Pilgrim 1 and Pilgrim 6 had similar (Dunn's $z = 0.59$, $p = 1$) rates of GPP (6.66 ± 2.11 and 6.48 ± 2.77 g O_2 m^{-2} d^{-1} mean ± standard deviation, here and throughout). GPP at both of these sites were significantly higher than those at 14 (mean rate of 2.09 ± 0.69 g O^2 m^{-2} d^{-1}; Pilgrim 1 vs. 14 $z = 10.99$, $p < 0.001$; Pilgrim 6 vs. 14 $z = 10.15$, $p < 0.001$). ER rates were highest downstream; Pilgrim 1 had the highest average ER of -10.46 ± 1.34 g O_2 m^{-2} d^{-1} compared to both Pilgrim 6 and Pilgrim 14 ($z = 5.02$, $p < 0.001$ and $z = 13.39$, $p < 0.001$, respectively). ER at Pilgrim 14 was -5.75 ± 0.99 g O_2 m^{-2} d^{-1}, which was significantly lower ($z = 7.96$, $p < 0.001$) than ER at Pilgrim 6 (-8.74 ± 1.03 g O_2 m^{-2} d^{-1}).

For the gradient study, there was no correlation between ER and k_{600} at any of the four study sites, suggesting no issues with GPP and ER model performance or equifinality (Online Supplementary Materials Fig. 7). Rates of GPP (Kruskal–Wallis, $\chi^2 = 54.77$, df = 3, $p < 0.001$) and ER ($\chi^2 = 126.74$, df = 3, $p < 0.005$) were significantly different among the four sites (Fig. 3). GPP at the Tobacco (1.10 ± 0.93 g O_2 m^{-2} d^{-1}) was approximately 3 × lower than the Trap Rock, Silver, and the Pilgrim 1 sites ($z = 6.453$, $z = 5.788$, and $z = 2.727$ respectively, $p < 0.05$ for all three). The Silver, Trap Rock, and Pilgrim 1 had similar rates of GPP (3.34 ± 2.75, 3.15 ± 2.68 and, 3.14 ± 4.73 g O_2 m^{-2}

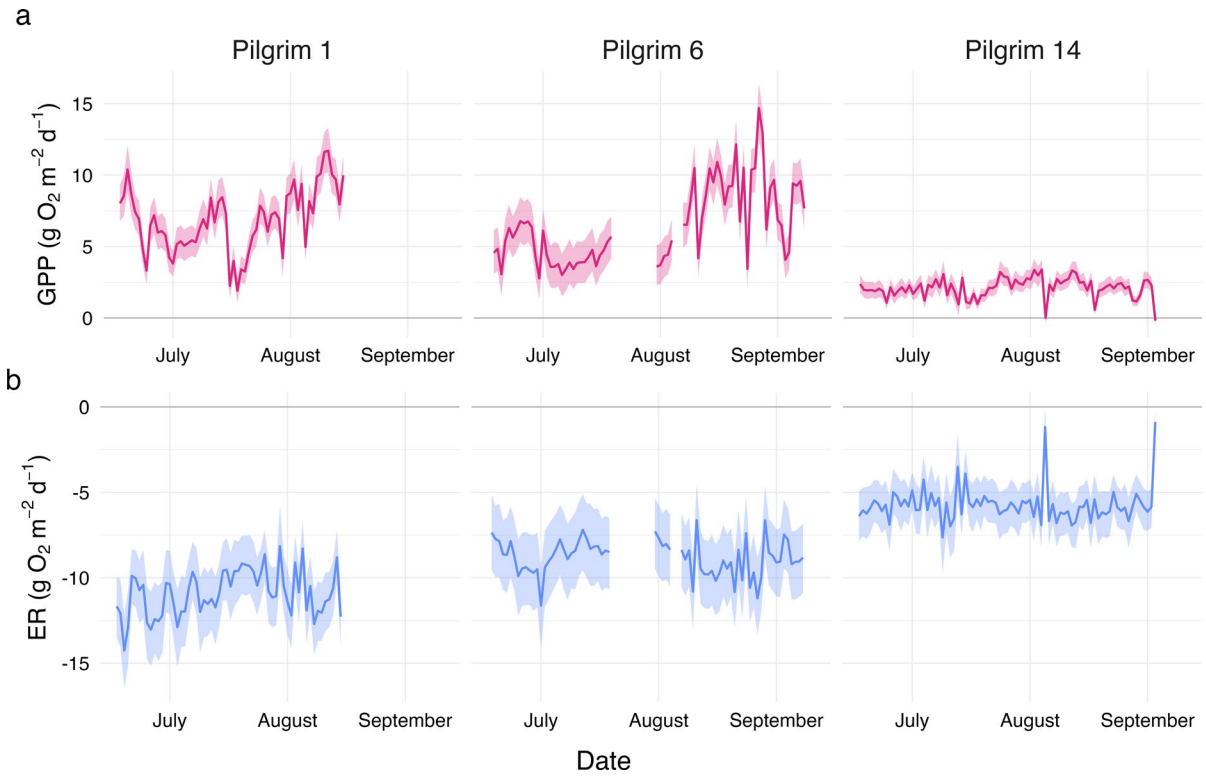

Fig. 2 Time series plots of metabolism estimates for Pilgrim 1 (left, downstream), Pilgrim 6, and Pilgrim 14 (right, upstream) during the Longitudinal Study in 2019. The pink lines are estimated GPP **a**, and blue is estimated ER **b**. The shaded ribbons represent the 95% confidence intervals. The gaps in data are due to sensor burial by sand

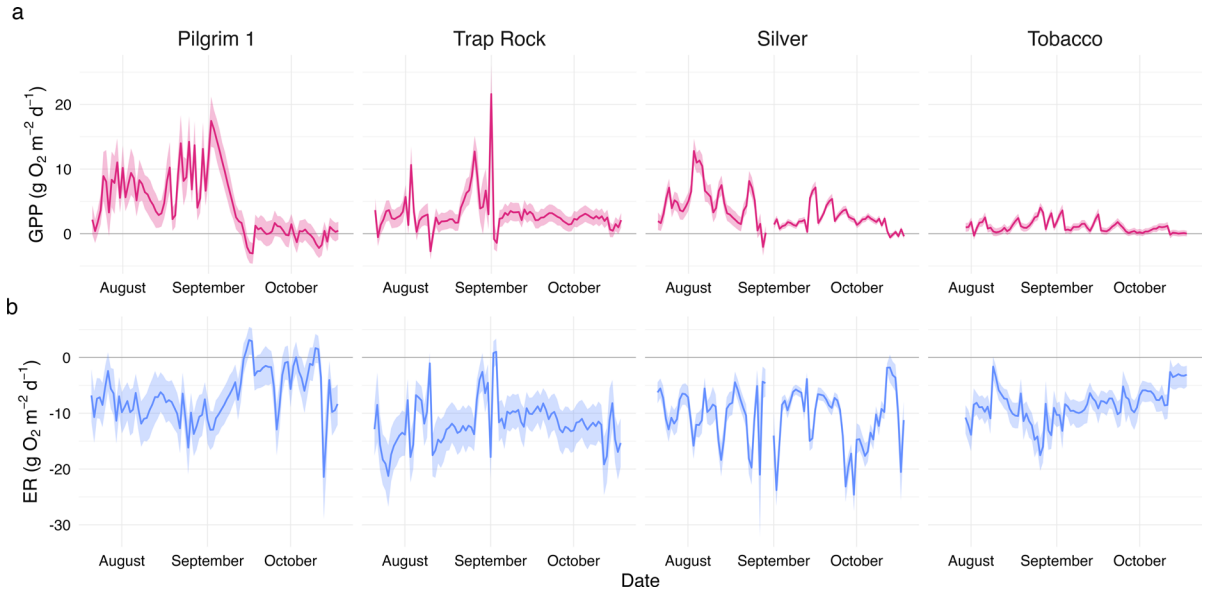

Fig. 3 Time series plots of **a** GPP and **b** ER estimates for the Pilgrim, Trap Rock, Silver, and Tobacco sites in the DOC Gradient Study in 2020. The lines are estimated GPP and ER with pink representing GPP and blue ER. Shaded ribbons represent the 95% confidence intervals

d^{-1}, respectively; Fig. 3a), however, the post hoc Dunn's test suggested that the Pilgrim 1 was significantly different from both the Silver ($z = 3.588$, $p = 0.002$) and the Trap Rock ($z = 4.208$, $p < 0.001$). For ER, the Trap Rock had the highest average rates (-12.2 ± 3.75 g O_2 m^{-2} d^{-1}) compared to the Silver, Tobacco, and Pilgrim River ($z = 3.489$, 5.907, and 10.988, respectively, $p < 0.05$ for all; Fig. 3b). Pilgrim 1 had lower average ER rates (-5.49 ± 4.5 g O_2 m^{-2} d^{-1}) than the Tobacco and Silver ($z = 3.973$ and $z = 6.669$, respectively, $p < 0.001$ for both), and the Silver (-10.7 ± 5.02 g O_2 m^{-2} d^{-1}) and the Tobacco were intermediate (-8.56 ± 3.75 g O_2 m^{-2} d^{-1}) and not significantly different from each other ($z = 2.355$, $p = 0.111$).

Autotrophic and heterotrophic respiration

For the Longitudinal Study, values of AR_f were estimated for the three sites where metabolism was estimated with variable success (Online Supplementary Materials Fig. 8). Two sites (Pilgrim 1 and Pilgrim 6) produced realistic estimates based on the criteria of Hall and Beaulieu (2013; $AR_f < 1$, $CI < 0.4$; Table 2), while Pilgrim 14 did not have enough variation within the rates of GPP to produce realistic estimates. Therefore, we were able to calculate AR and HR for only Pilgrim 1 and 6. AR at Pilgrim 1 was higher (-3.79 ± 1.58 g O_2 m^{-2} d^{-1}; Fig. 4a) than AR at Pilgrim 6 (-2.35 ± 1.18 g O_2 m^{-2} d^{-1}; $z = 4.380$, $p < 0.001$). In contrast, HR did not differ between Pilgrim 6 (Fig. 4b, -6.39 ± 1.01 g O_2 m^{-2} d^{-1}) and Pilgrim 1 (-6.67 ± 1.71 g O_2 m^{-2} d^{-1}; $z = 0.691$, $p = 1$).

For the DOC Gradient Study, realistic estimates of AR_f were produced for only two of the four study sites (Pilgrim 1, Silver; Table 2). The Tobacco did not have enough variation in GPP to produce realistic estimates of AR_f (Online Supplementary Materials Fig. 9). HR in the Trap Rock was correlated with GPP ($r = 0.52$), causing AR_f to have a CI greater than 0.4, not meeting the criteria for inclusion in further analyses. The Silver had slightly but significantly higher rates of AR (-2.74 ± 2.01 g O_2 m^{-2} d^{-1}) than Pilgrim 1 (-2.09 ± 3.33 g O_2 m^{-2} d^{-1}, $z = 2.996$, $p = 0.016$, Fig. 4a), but had much higher rates of HR (Silver: -9.04 ± 5.51 g O_2 m^{-2} d^{-1}, Pilgrim 1: -3.28 ± 3.14 g O_2 m^{-2} d^{-1}; $z = 10.338$, $p < 0.001$, Fig. 4b).

Decomposition

Decomposition rates (k_d) did not differ between Longitudinal Study sites (ANOVA, $F = 2.4$, $df = 1,23$, $p = 0.13$). However, decomposition rates corrected for degree days (k_{dd}) were different among some sites (ANOVA, $F = 16.9$, $df = 4,20$, $p < 0.001$) but were not lower at downstream sites as we predicted. Pilgrim 15 had the lowest average temperature-corrected rate ($3.27 \times 10^{-4} \pm 2.67 \times 10^{-4}$ k_{dd}) and was the only site that differed from the others based on the *post hoc* Tukey HSD test ($p < 0.001$ for Pilgrim 15 vs. all other sites). There were no significant differences in decomposition rates among the other 4 sites (Fig. 5b).

There were no significant differences in k_d (ANOVA, $F = 1.18$, $df = 3,12$, $p = 0.36$) or k_{dd} across DOC Gradient Study sites (ANOVA, $F = 1.16$, $df = 3,12$, $p = 0.36$). Although not significant, Pilgrim

Table 2 Estimates of the fraction of gross primary production that fuels autotrophic respiration (AR_f) in Longitudinal and DOC Gradient study sites

	Longitudinal Study			DOC Gradient Study			
Site	Pilgrim 14	Pilgrim 6	Pilgrim 1	Pilgrim 1	Tobacco	Silver	Trap Rock
AR_f	− 1.33	− 0.39	− 0.62	− 0.70	− 2.52	− 0.70	− 0.51
2.5 pct	− 1.69	− 0.44	− 0.85	− 0.76	− 2.83	− 0.78	− 0.56
97.5 pct	− 1.12	− 0.37	− 0.48	− 0.68	− 2.09	− 0.58	0.09
95% CI	0.57	**0.07**	**0.37**	**0.08**	0.74	**0.20**	0.65

AR_f was calculated as the slope of a quantile regression of gross primary production and ecosystem respiration using the 0.9 quantile. AR_f estimates are unreliable if the 95% confidence interval (CI), calculated as the difference between the 97.5th percentile of the AR_f estimate (97.5 pct) and the 2.5th percentile of the estimate (2.5 pct), is greater than 0.4. Sites with a 95% CI less than 0.4 are indicated in bold

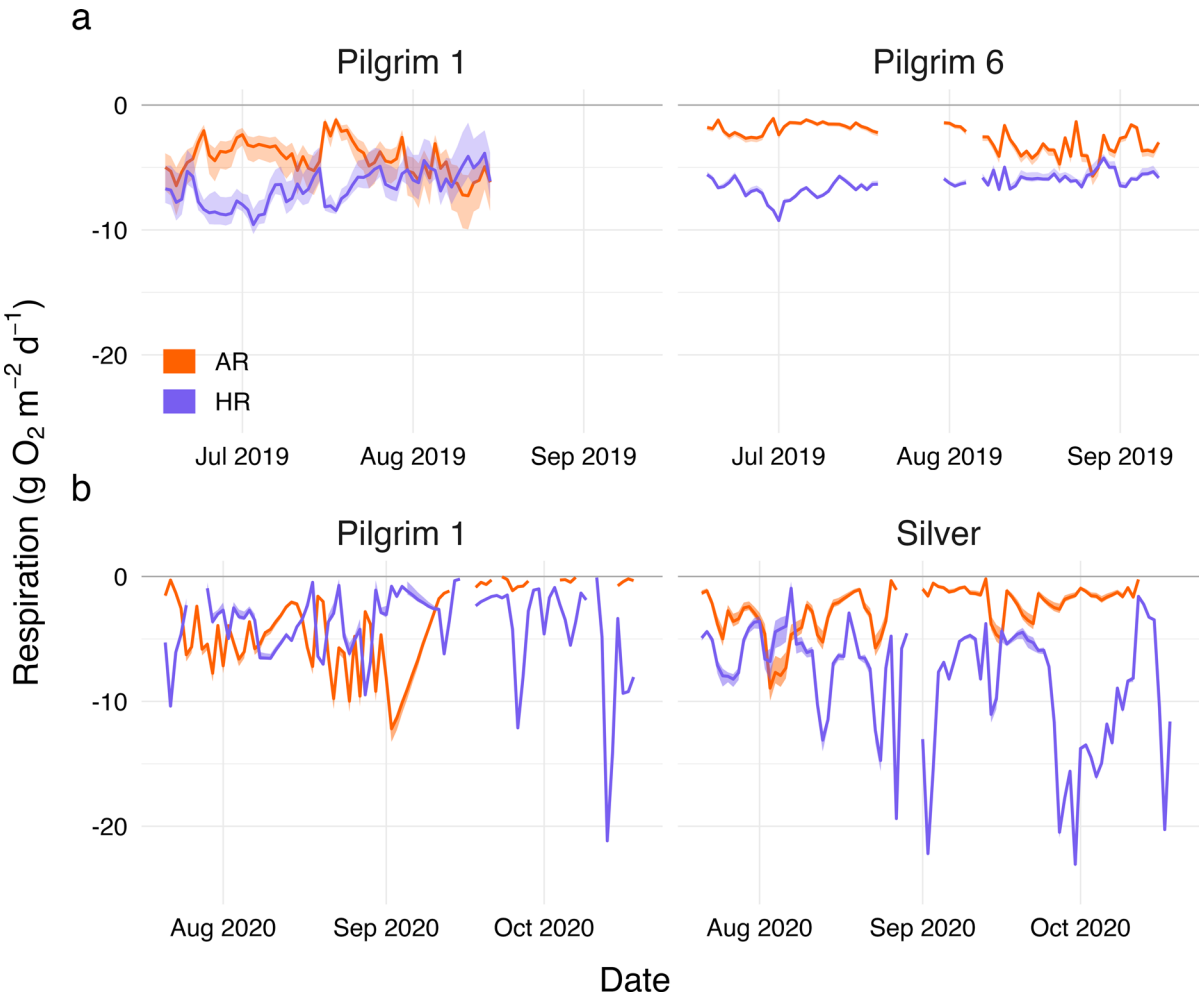

Fig. 4 Time series plots of AR and HR rates at the four sites where the values could be calculated. The solid line indicates respiration rates while the shaded ribbons around each represent the 95% confidence intervals. **a** shows Pilgrim 1 and Pilgrim 6 from the 2019 Longitudinal Study, **b** shows Pilgrim 1 and Silver from the 2020 DOC Gradient Study

1 and the Silver had the highest k_{dd} among these sites ($1.07 \times 10^{-3} \pm 5.95 \times 10^{-4}$ and $1.31 \times 10^{-3} \pm 8.92 \times 10^{-4}$ respectively; Fig. 5b).

Environmental factors versus process rates

Partial least squares regression (Fig. 6) and correlation analyses (Online Supplementary Materials Fig. 10) showed that nutrient concentrations and canopy cover were important predictors of ER and decomposition rates, and DOC was not an important predictor of any of the processes we measured. For ER, the PLSR performed best using the first two components and explained 32.00% of the variance in rates (Fig. 6a). Only TDP was identified as an important predictor of ER (2.29 VIP), with all other factors having VIP scores < 1 (Table 3). For decomposition, the PLSR analysis for rates of k_{dd} best performed with the first 4 components and explained 97.59% of the variance in rates (Fig. 6b). While this model had no highly influential predictors, NO_3^- (1.61 VIP), TDP (1.50 VIP), and canopy cover (1.19 VIP) were all moderately influential predictors (Table 3).

Because we could only estimate AR and HR at 4 sites, we limited the analysis of rates versus environmental variables to correlations (Online Supplementary Materials Fig. 10). Rates of AR were not significantly correlated with any environmental predictor

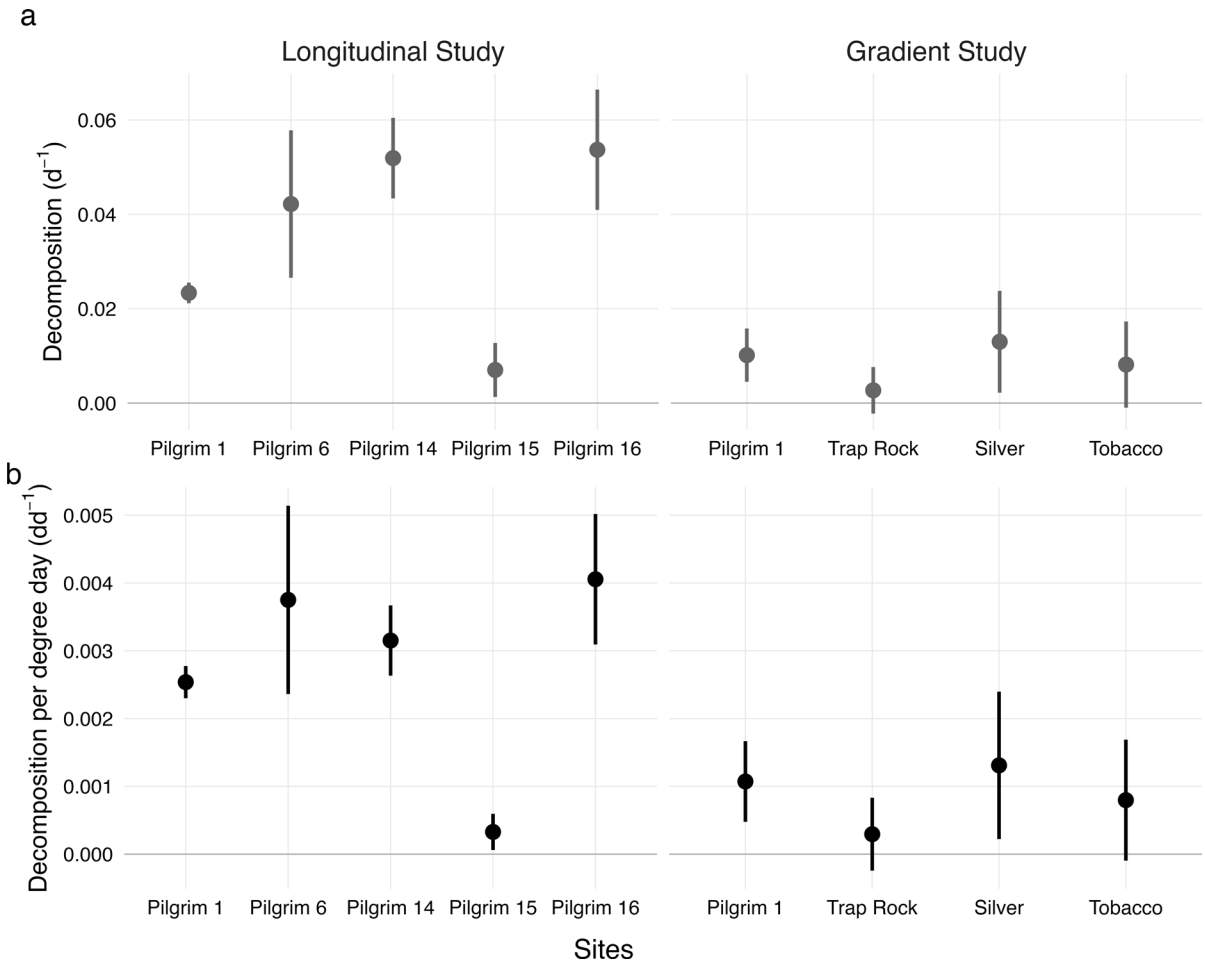

Fig. 5 Comparison of **a** decomposition rate per day and **b** decomposition rate per degree day in sites from the 2019 Longitudinal and 2020 DOC Gradient studies. Points denote means, error bars denote standard deviation

but were highly correlated with rates of GPP ($r = 0.98$, $p < 0.0001$), which is an artifact of how AR rates were calculated (see methods above). Rates of HR were not significantly correlated with any environmental characteristics, and were strongly correlated with ER ($r = 0.72$, $P < 0.001$), which is also a calculation artifact. In contrast with our predictions, neither AR nor HR were significantly correlated with DOC concentrations, nor was HR correlated with canopy cover.

Discussion

Relationships between carbon cycling processes and environmental predictors

For both the Longitudinal and DOC Gradient studies, we hypothesized that rates of ER and AR would respond to changes in canopy cover or DOC due to their effects on light. Consistent with the first hypothesis in the Longitudinal Study, we found that ER was greatest at the most downstream site. However, when we combined data from both studies, we found that ER and AR were not correlated with canopy cover, nor was canopy cover identified as an important predictor of ER in the PLSR analysis. Consistent with the hypotheses for the DOC

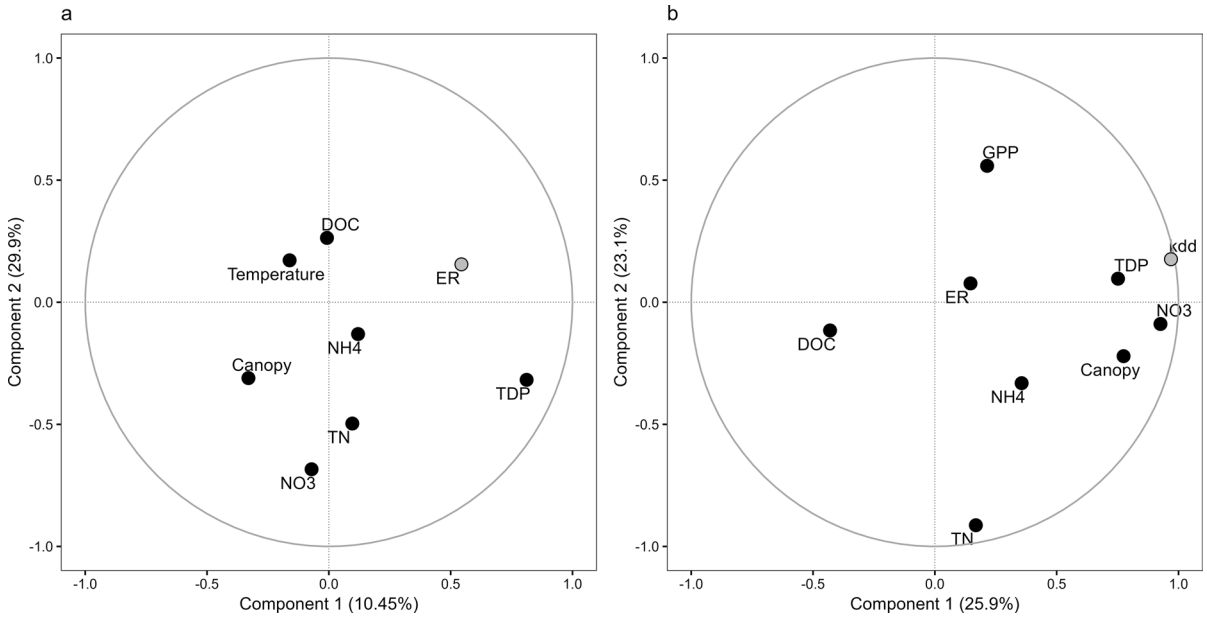

Fig. 6 Partial least squares loading plots for **a** ecosystem respiration (ER) and **b** decomposition rates per degree day (k_{dd}) with predictor variables canopy cover (Canopy), dissolved organic carbon (DOC), gross primary productivity (GPP), ammonium (NH4), nitrate (NO3), total dissolved phosphorus (TDP), temperature, and total dissolved nitrogen (TN)

Gradient Study, DOC was not an important predictor for ER. For example, the Tobacco site, which had much higher DOC concentrations relative to the other study sites, did not have ER rates that were significantly different from the other sites. When we combined data from both studies, DOC concentrations were not related to ER, AR, or HR rates across sites. This contradicts our prediction of an offset in rates of ER with increasing DOC concentrations, with AR decreasing and HR increasing as DOC increased. Combined, the two studies we conducted provide limited support for our hypotheses about canopy cover on ER, but no support for our hypotheses about DOC effects on stream respiration. It is important to note that we did not measure light directly, and we may have found stronger relationships if we had not relied on indirect drivers of light availability.

Table 3 Variable importance on the projection (VIP) scores for the partial least squares regression for ecosystem respiration (ER) and decomposition rate per degree day (k_{dd})

Predictor Variable	ER VIP Score	k_{dd} VIP Score
Dissolved organic C	0.30	0.77
Canopy Cover	0.87	**1.19**
TN	0.47	0.40
TDP	**2.29**	**1.50**
NH4	0.30	0.69
NO3	0.69	**1.61**
GPP	*	0.61
ER	*	0.42
Temperature	0.37	*

Asterisks indicate predictor variables that were not included in the analysis because they were directly used in the modeling or calculation of the response variable

Our second hypothesis was that canopy cover would be a significant predictor of decomposition and ER due to changes in supply of allochthonous materials. Decomposition rates in these streams (0.003–0.054 k_d) were within the range of rates reported in northern forested rivers in a global cross-site comparative study (0–0.2 k_d; Tiegs et al., 2019), and very similar to those reported in other Upper Peninsula streams (0.002–0.094 k_d; Mancuso et al., 2022). One of the sites with the lowest rates of decomposition was Pilgrim 15. Although this is a headwater site, it was in a beaver and wetland complex with an open canopy. In contrast, the sites with the highest decomposition rates, Pilgrim 14 and Pilgrim 16, had greater than 90% canopy cover. The fact

that we observed higher decomposition rates at sites with higher canopy cover may be due to these sites receiving more allochthonous inputs of terrestrial detritus. This higher supply of allochthonous material could sustain a larger population of cellulose decomposers (bacterial, fungal, and invertebrate; Aguiar et al., 2017; Tonin et al., 2018), which in turn would allow for more rapid colonization and decomposition of the cotton strips. Alternatively, these differences in rates could potentially be due to a negative priming effect of algae on the heterotrophs (Halvorson et al., 2019). The lower density of canopy cover could lead to higher primary production, which in turn can lead to a greater supply of autochthonous material. This greater abundance of autochthonous material could be more labile than the cotton that was provided during this experiment, leading to lower measured decomposition rates due to the heterotrophs preferring the autochthonous material.

Finally, we evaluated whether environmental conditions like water temperatures and nutrient concentrations would be related to rates of ER, AR, HR, and decomposition. We found that temperature was not significantly correlated with ER and AR rates, nor was temperature identified as an important predictor of ER in the PLSR analysis. The lack of strong relationships between temperature and other processes may be due to the low variation in water temperature (12.7–16.9 °C) across sites during the study period. Alternatively, this lack of correlation could be due to these studies encompassing multiple seasons. Figure 4 shows that the rates of HR in late October for the sites in the DOC Gradient Study were some of the highest in our time series, despite that this is a time when temperature decreases. October is normally the time of leaf fall, so decreased temperatures may be offset by an increase in organic matter subsidies that can fuel much more heterotrophic activity (e.g., Roberts et al., 2007; Coble et al., 2019), which may obfuscate relationships between temperature, ER, and HR. Unfortunately, the metabolism time series from the current study are too short to conduct a seasonal analysis of how temperature and other environmental drivers may interact to control metabolism and decomposition rates.

The most important set of environmental predictors of ER and decomposition rates were nutrient concentrations. PLSR analyses identified TDP as a very important predictor of ER and a moderately important predictor of k_{dd} rates, and NO_3^- was also a moderately important predictor of decomposition rates. Nutrient concentrations were expected to be top contributors to these rates as previous studies have already established that nutrient availability is important for many in-stream processes (Griffiths et al., 2013; Rosemond et al., 2015). These relationships between process rates and nutrient concentrations may be due to certain forms of nitrogen or phosphorus being limiting to organisms in the streams (Reisinger et al., 2016), and other studies have found that higher nutrient concentrations lead to higher rates of decomposition (Benstead et al., 2009; Ferreira et al., 2015). Organic matter is generally nutrient-poor, and microbes that decompose organic matter often need to obtain nutrients from the environment to use the carbon in organic matter and build their biomass (Suberkropp & Chauvet, 1995). This is particularly true for the decomposition assays we conducted, as the material used for the standardized cotton strip assay is low in nutrients, which requires that microbes immobilize nutrients to decompose the test materials (Costello et al., 2022).

Limitations to estimates of AR and HR

It is important to consider the limitations of the quantile regression method, which cannot resolve AR and HR in all streams (Hall & Beaulieu, 2013). Using this method, we were able to statistically separate AR and HR at only four of the eight sites. Streams with low GPP variability ($AR_f < 1$, $CI < 0.4$) are unsuitable for the quantile regression approach (Hall & Beaulieu, 2013). This is the most probable reason that we were unable to estimate HR and AR for the blackwater Tobacco. Typically, blackwater rivers have lower than average rates of GPP in comparison to non-blackwater rivers, with higher rates of ER in blackwater rivers than in other rivers with comparable productivity (Meyer & Edwards, 1990). Since comparatively little primary production occurred at this site (Fig. 4a), it could be inferred that HR primarily contributed to ER.

This study is one of a small set that have applied the statistical approach developed by Hall & Beaulieu (2013) to partition AR and HR rates (e.g., Griffiths et al., 2013; Ulseth et al., 2018; Arroita et al., 2019; Hosen et al., 2019; summarized in Table 4). Hall & Beaulieu (2013) estimated rates for 13 different

Table 4 Summary of studies that have quantified AR_f values that used the methods of Hall & Beaulieu 2013, including the current study

Location	Stream Name	Year	AR_f	Study
Basque Country, Spain	Berriatua	2003–2004	0.11	Hall & Beaulieu, (2013)
Alegia, Spain	Oria River	2016	0.15	Arroita et al., (2019)
Basque Country, Spain	Erenozu	2003–2004	0.17	Hall & Beaulieu, (2013)
Basque Country, Spain	Oiartzun	2003–2004	0.30	Hall & Beaulieu, (2013)
Idaho, USA	Portneuf River	2003–2008	0.34	Hall & Beaulieu, (2013)
Connecticut, USA	Sleeper's River	2015–2017	0.34	Hosen et al., (2019)
Basque Country, Spain	Gardea	2003–2004	0.35	Hall & Beaulieu, (2013)
West of Vienna, Austria	Ybbs River Network	2012–2014	0.39	Ulseth et al., (2018)
Michigan, USA	Pilgrim 6	2019	0.39	Current Study
Basque Country, Spain	Nuxika	2003–2004	0.41	Hall & Beaulieu, (2013)
Tennessee, USA	Walker Branch	2004–2006	0.43	Hall & Beaulieu, (2013)
Basque Country, Spain	Aizarnazabal	2003–2004	0.47	Hall & Beaulieu, (2013)
Basque Country, Spain	Lasarte	2003–2004	0.47	Hall & Beaulieu, (2013)
Connecticut, USA	Farmington River	2015–2017	0.48	Hosen et al., (2019)
Connecticut, USA	Connecticut River	2015–2017	0.49	Hosen et al., (2019)
Connecticut, USA	Passumpsic River	2015–2017	0.59	Hosen et al., (2019)
Michigan, USA	Pilgrim 1	2019	0.62	Current Study
Basque Country, Spain	Aitzu	2003–2004	0.64	Hall & Beaulieu, (2013)
Basque Country, Spain	Alegia	2003–2004	0.64	Hall & Beaulieu, (2013)
Basque Country, Spain	Anorebieta	2003–2004	0.65	Hall & Beaulieu, (2013)
Connecticut, USA	Still River	2015–2017	0.67	Hosen et al., (2019)
Ohio, USA	Shepherd Creek	2009–2011	0.69	Hall & Beaulieu, (2013)
Indiana, USA	Shatto Creek	2007	0.70	Griffiths et al., (2013)
Michigan, USA	Pilgrim 1	2020	0.70	Current Study
Michigan, USA	Silver	2020	0.70	Current Study
Alegia, Spain	Oria River	2004	0.81	Arroita et al., (2019)
Alegia, Spain	Oria River	2003	1.01	Arroita et al., (2019)

The stream names as given by the original study; location (city or state, country) and years of the studies are also given for comparison. The table is arranged from lowest to highest AR_f values, moving from top to bottom

streams from Idaho, Ohio, Tennessee, and Northern Spain. While the rates of GPP in the current study were similar to sites used by Hall & Beaulieu (Roberts et al., 2007; Izagirre et al., 2008; Marcarelli et al., 2010; Beaulieu et al., 2013), the four sites where we were able to estimate AR_f had a much greater proportion of primary production being used to fuel AR (Table 4). Three of the study streams had AR_f values between 0.62 and 0.70, corresponding to the mid- to high-end of rates published in literature. The largest AR_f value that has been reported in the literature was 1.01 in the Oria River in Alegia, Spain (Arroita et al., 2019); however, that was in 2003 before the installation of a wastewater treatment plant in the

area, and in 2016 that same river had an AR_f value of 0.15 (Table 4), so the AR_f may have been fueled by the eutrophic conditions that were present. In contrast, our sites have relatively low nutrient availability and streams in the region exhibit high demand for dissolved nutrients characteristic of nutrient limitation (Coble et al., 2016a). Because AR_f estimates from these sites in the current study and other studies show a wide variation within small geographic regions, we hypothesize that there are unconsidered environmental or biological factors that influence the proportion of GPP that supports AR. This study highlights the need for a more robust method for partitioning AR and HR to improve understanding of the environmental and biological factors that differentially control rates of respiration, and the partitioning of GPP to respiration by autotrophs.

Insights on carbon cycling in northern forested streams

While our results did not support our hypotheses about direct relationships between DOC concentration and canopy cover on ER, AR, and HR, our results still provide valuable insights into the carbon cycle of northern forested streams and how they may respond to anthropogenic change. The blackwater Tobacco had the highest concentrations of DOC and the lowest rates of GPP, likely due to light limitation. Yet, this site had comparable ER rates to those measured in the other study rivers, which suggests that as DOC concentrations increase, streams can continue to remove DOC through heterotrophic respiration. This will be important to consider as many DOC concentrations are increasing in forested streams globally, including Lake Superior tributaries (Roulet & Moore, 2006; de Wit et al., 2016; Meingast et al., 2020). Additionally, while our results did not show that temperature was an important predictor of decomposition and respiration, it is well known that temperature is a direct control of these processes (Demars et al., 2011; Fernandes et al., 2014). This lack of relationship may be due to the small number of streams included in the study, and the small temperature variation during the study period and between the study sites. It will be important to understand how these processes vary throughout the seasons and in relation to key seasonal events, such as timing of water, C, and nutrient pulses into rivers. Understanding the controlling factors of respiration and decomposition is crucial to better integrate inland aquatic ecosystems into global climate models.

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Author contributions RCS and AMM contributed to the study conception and design. Data collection was led by RCN, and data analyses were performed by RCN and MCK. The first draft of the manuscript was written by RCS and all authors commented on the previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability All data used in this study are available through the Environmental Data Initiative: Schipper, R.C., M.C. Kelly, and A.M. Marcarelli., 2023. Metabolism and decomposition rates from 5 Lake Superior Tributaries, 2018–2019 ver 1. Environmental Data Initiative. https://doi.org/https://doi.org/10.6073/pasta/dfc8a2319bf8b8a6269bfd94895df873.

Declarations

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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